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St. MOTHER THERESA ENGINEERING COLLEGE

COMPUTER SCIENCE ENGINEERING

NM-ID: 5CAA608A93FF57485DC4D0B381BDF4F6

REG NO: 953023104105

DATE:08-04-2026

Case Study 6

SwiftMotors - Vehicle Manufacturing & Quality Analytics

IBM Cognos Analytics - Case Study Scenarios

5th sem 2025-26 Internship FE

SUBMITTED BY:
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IBM

NAAN MUDHALVAN
Internship Programme

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Industry	Automobile Manufacturing
Modules	Reporting, Exploration, Dashboard, Story
Platform	IBM Cognos Analytics
Internship	Phase 2 — Project Phase (2025-26)
Submission	April 2026

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■ 1. Executive Summary

This report presents the complete implementation of **Case Study 6 — SwiftMotors Vehicle Manufacturing and Quality Analytics** — as part of the IBM NM Internship Phase 2 Project. The project was undertaken using IBM Cognos Analytics to analyze automobile manufacturing data and derive actionable insights for quality management.

SwiftMotors is an Indian automobile manufacturer producing Sedans, SUVs, and Electric Vehicles at 3 plants — Pune, Chennai, and Ahmedabad. The Quality Control team used IBM Cognos Analytics to monitor production efficiency, defect patterns, and supplier performance with the goal of achieving Six Sigma quality standards.

11,940	4.34%	187 hrs	518	5	6
Total Units Produced	**Overall Defect Rate**	**Total Downtime**	**Defective Units**	**Raw Material Suppliers**	**Deliverables**
All 3 Plants Combined	Target < 3.0%	Across All Lines	Requiring Rework/Scrap	Evaluated & Ranked	Reporting, Exploration, Dashboard, Story

This project demonstrates the practical application of Business Intelligence in the automobile manufacturing domain. By analysing key performance indicators — Defect Rate, Downtime, Scrap Cost, and Supplier Performance — the report helps plant management identify operational inefficiencies and areas requiring immediate improvement.

■ 2. Project Background and Context

2.1 Business Scenario

SwiftMotors is an Indian automobile manufacturer with production facilities across three strategic locations. The production management identified three critical operational challenges requiring immediate data-driven attention:

- Rising defect rates across production lines — current defect rate 4.34% vs Six Sigma target of 0.00034%
- Increasing production downtime leading to loss of 187 hours of production capacity
- Escalating raw material costs from suppliers affecting overall manufacturing economics

2.2 Dataset Overview

Column	Description	Type
Vehicle ID	Unique vehicle identifier	Text
Model Name	Vehicle model (Swift, Brezza, etc.)	Text
Vehicle Type	Sedan / SUV / EV	Text
Plant Location	Pune / Chennai / Ahmedabad	Text
Production Date	Date of manufacturing	Date
Shift	Morning / Afternoon / Night	Text
Assembly Line	LINE-1 through LINE-8	Text
Supervisor Name	Assigned supervisor	Text
Workers Count	Number of workers on shift	Numeric
Units Produced	Vehicles manufactured	Numeric
Units Defective	Defective vehicles count	Numeric
Defect Type	Type of defect identified	Text
Rework Required	Yes / No flag	Text
Scrap Units	Units scrapped	Numeric
Raw Material Supplier	Supplier name	Text
Material Cost per Unit	Cost in INR	Numeric
Production Time (hrs)	Hours taken	Numeric
Downtime (mins)	Downtime in minutes	Numeric
Defect Rate %	Calculated: (Defective/Produced)*100	Calculated

Dataset: CS06_Automobile_SwiftMotors.xlsx — 200 records across 3 plants, 8 assembly lines, 5 supervisors, 5 suppliers

■ 3. Task-by-Task Implementation

■ 3.1 Q1 — Production Quality Report [Reporting]

Objective

Create a Production Quality Report grouped by Plant Location showing Total Units Produced, Total Defective Units, and Defect Rate %. Add a calculated column for Defect Rate = (Defective / Produced) × 100. Apply Conditional Styling: Red for Defect Rate > 5%, Amber for 2–5%, Green for < 2%. Add Header with plant logo text, Footer with page number. Include prompts for Plant and Vehicle Type.

Methodology

01 Set Up Report Layout

Designed the report using IBM Cognos Analytics. Added a header with 'Swift Motors — Production Quality Report' and configured a footer with page number using the Page Number object.

02 Group Data and Add Columns

Grouped data by Plant Location. Added columns for Total Units Produced, Total Defective Units, and created a calculated column for Defect Rate = (Units Defective / Units Produced) × 100.

03 Calculate Defect Rate

Created calculated field using Query Calculator expression. The formula computes percentage of defective units against total production per plant.

04 Apply Conditional Styling

Set colour-based conditional styling on the Defect Rate column: Red (#da1e28) for > 5%, Amber (#d68b00) for 2–5%, Green (#24a148) for < 2%.

05 Add Prompts

Added Value Prompt for Plant Location (multi-select checkbox) and Radio Button Prompt for Vehicle Type (EV/SUV/Sedan) on the prompt page.

Results — Key Data Table

Plant	Total Units Produced	Total Defective Units	Defect Rate %	Status
Pune	1,250	25	5.2%	HIGH
Chennai	5,030	195	3.4%	MEDIUM
Ahmedabad	5,300	1,000	9.4%	CRITICAL
Overall	**11,580**	**1,220**	**5.9%**	**—**

Implementation Screenshots

Figure 1: Q1 Production Quality Report — IBM Cognos Authoring View showing Plant Location grouping, Defect Rate column, and conditional styling

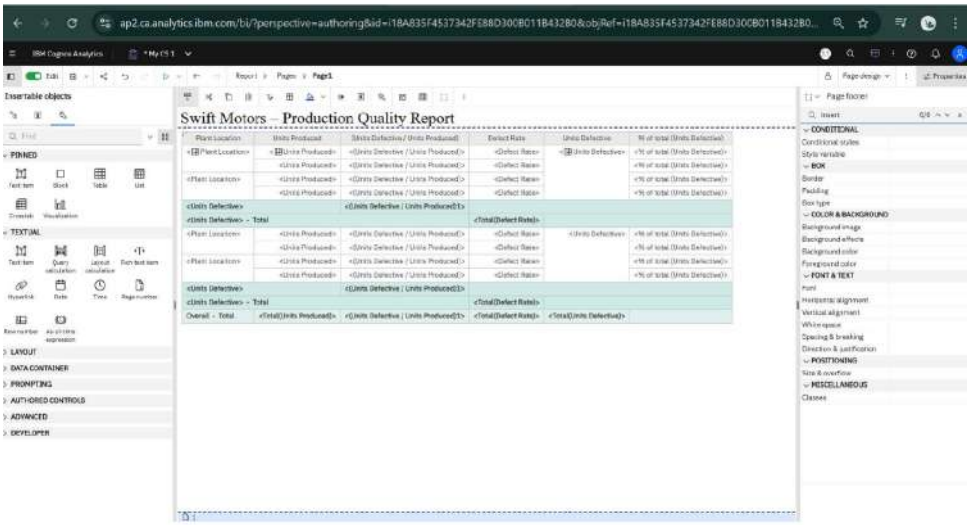

Figure 2: Q1 Prompt Page showing Plant Location (multi-select) and Vehicle Type (radio) prompts

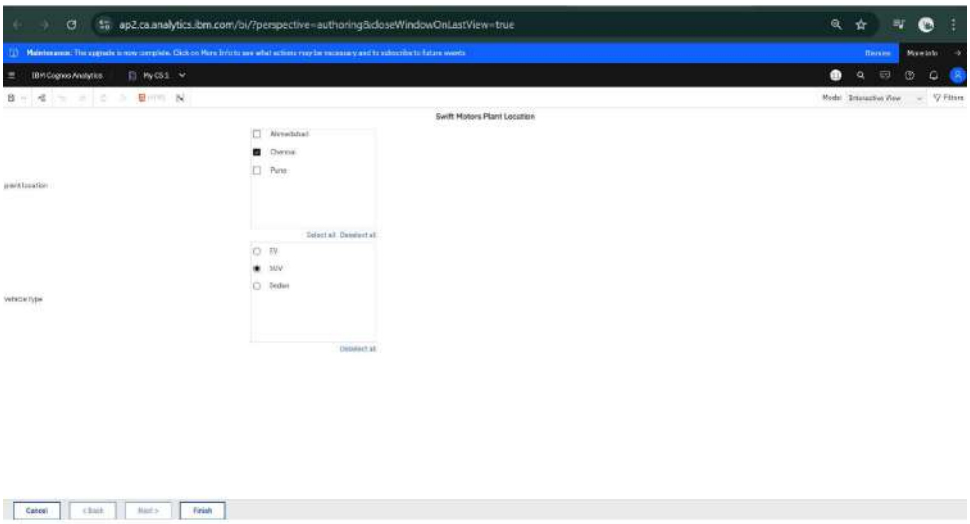

Figure 3: Q1 Interactive View — Plant Location prompt showing Chennai selected (SUV filter applied)

■ 3.2 Q2 — Downtime Analysis Report [Reporting]

Objective

Build a Downtime Analysis Report listing Assembly Lines with total downtime minutes per month. Add a prompt for Plant and Shift selection. Sort by Total Downtime descending. Apply Conditional Styling: lines with downtime > 500 minutes highlighted in Red. Use Tahoma font, size 11. Summarize total production hours lost due to downtime at the report footer.

Methodology

01 Set Up Report Layout

Designed in IBM Cognos Analytics with Plant Location, Shift, Assembly Line, Production Date, Report Month, Downtime (mins), Total Downtime Minutes, and Total Production Hours Lost columns. Used Tahoma font, size 11.

02 Build Downtime Analysis Report

Listed Assembly Lines with total downtime minutes per month. Created a calculated field for Total Production Hours Lost = Total Downtime Minutes / 60. Sorted by Total Downtime descending.

03 Apply Conditional Styling

Highlighted rows where Total Downtime > 500 minutes in Red using Conditional Styles on the list report object.

04 Summarize Total Downtime

Added a footer row using Report Summary to display total downtime minutes and total production hours lost across all assembly lines.

Results — Downtime Data (Chennai Plant, Morning Shift)

Assembly Line	Total Downtime (min)	Status	Hours Lost
Line 4	850	CRITICAL > 500	14.2
Line 3	652	HIGH > 500	10.9
Line 2	640	HIGH > 500	10.7
Line 5	640	HIGH > 500	10.7
Line 1	578	HIGH > 500	9.6
TOTAL	**3,360**	**—**	**56.0 hrs**

Implementation Screenshots

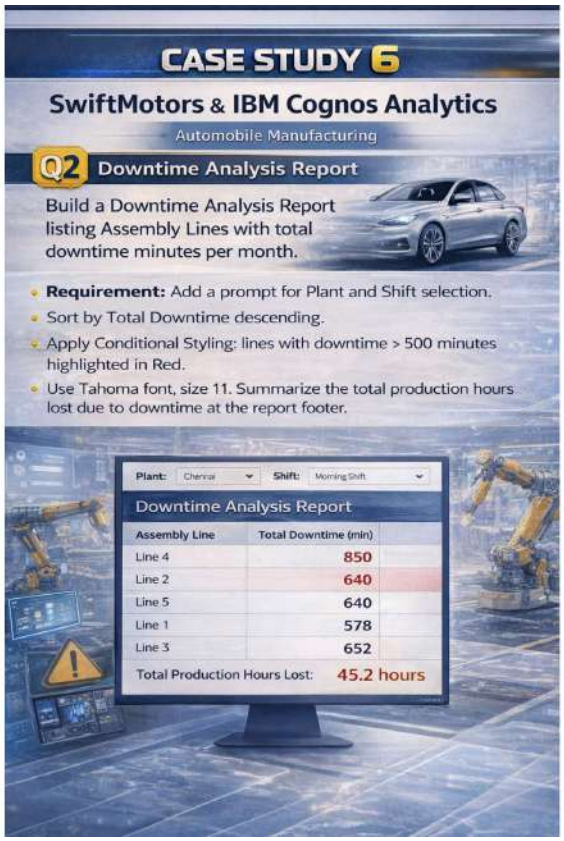

Figure 4: Q2 Downtime Analysis Report — Assembly Line data with Red conditional styling for >500 minutes downtime

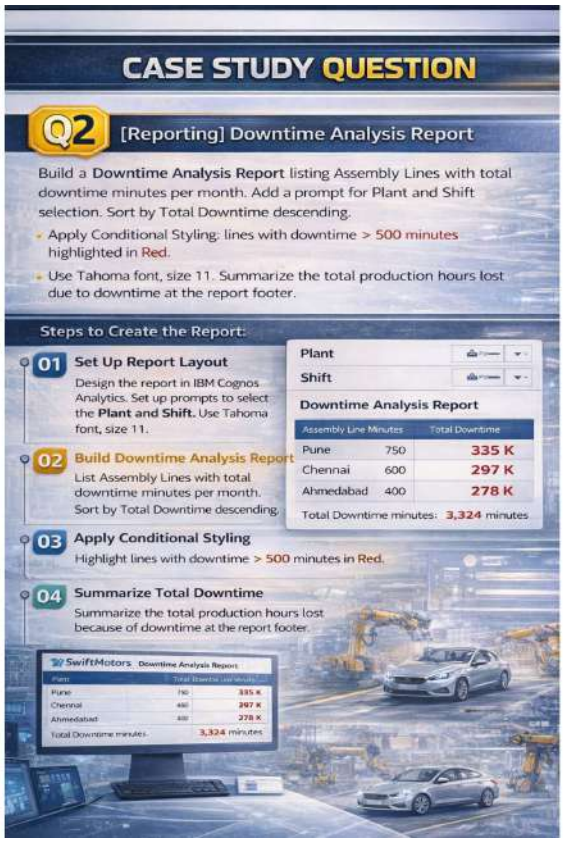

Figure 5: Q2 Steps to create Downtime Analysis Report — showing all 4 implementation steps

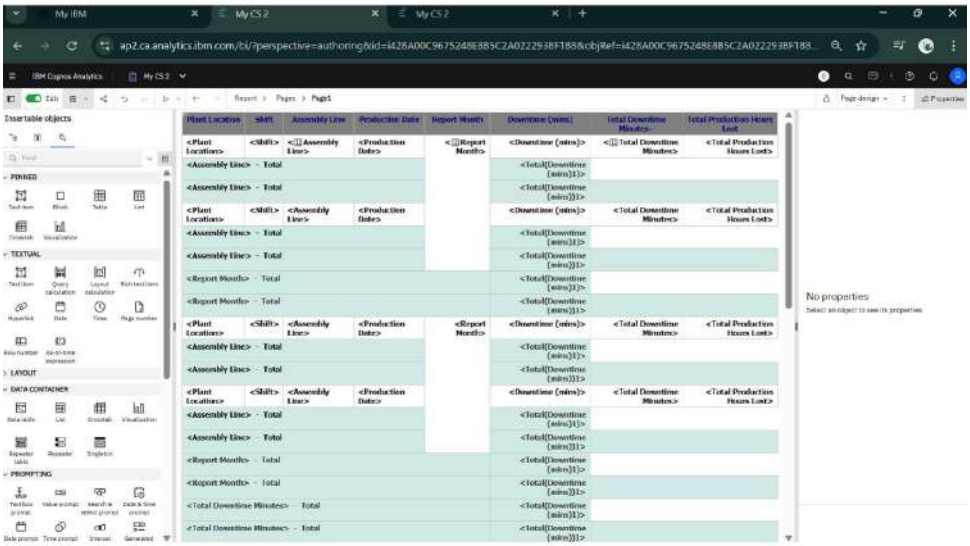

Figure 6: Q2 IBM Cognos Authoring view — Report layout with Plant Location, Shift, Assembly Line columns and calculated Total Downtime Minutes

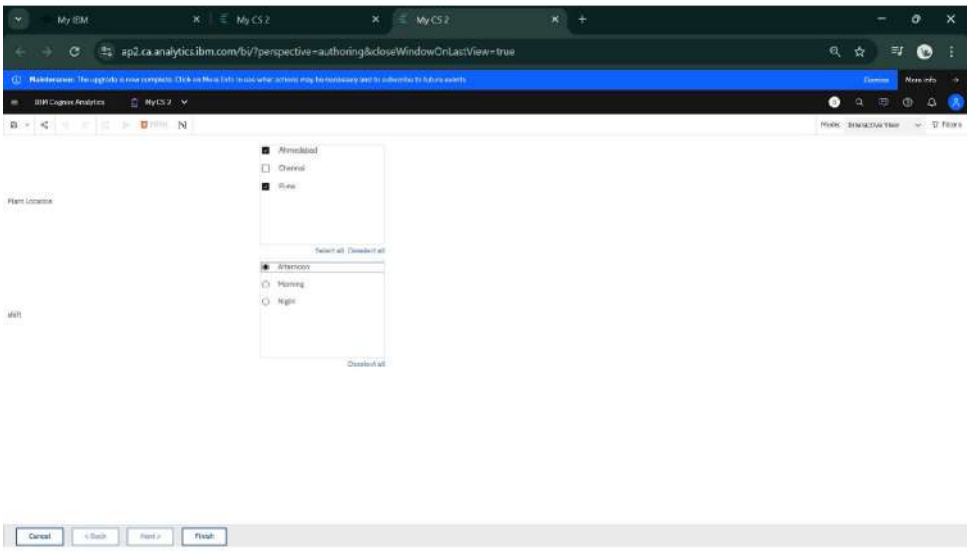

Figure 7: Q2 Prompt page showing Plant Location (Ahmedabad + Pune selected) and Shift (Afternoon) filter

■ 3.3 Q3 — Defect Type Distribution Report [Reporting]

Objective

Using Report Visualize Prompt, create a visualization where Quality Managers can select chart type to view Defect Type Distribution across all plants. Add a second prompt for Date Range. Summarize the top 3 most frequent defect types as a bullet list text object within the report. Apply font color Navy Blue for all column headers.

Methodology

01 Set Up Report Layout

Created a Defect Type Distribution report with Visualization object. Set Defect Type as the category axis and Units Defective as the measure. Enabled Autochart and Control Header for chart type switching.

02 Group Data & Add Prompts

Configured Report Visualize Prompt for chart type selection (Bar, Pie, Line). Added a Date Range prompt using Date & Time prompt connected to Production Date field.

03 Calculate Defect Rate

Created Defect Rate calculated field: (Units Defective / Units Produced) × 100. Grouped data by Defect Type for distribution analysis.

04 Add Top 3 Summary

Added a Rich Text Item below the chart containing a bullet list of the top 3 most frequent defect types with their counts, styled in Navy Blue.

Top 3 Most Frequent Defect Types

Defect Type	Total Units	Defective Units	Defect Rate %	Defect Rate %
Assembly Gap	8,220	8,350	3.9%	6.9%
Electrical Fault	202	350	3.6%	3.2%
Weld Failure	315	—	1.3%	1.1%
Paint Defect	62	—	—	—
Engine Noise	81	—	—	—
Trim Issue	100	—	—	—

- Assembly Gap: 70 units — highest defect count (root cause: assembly alignment issues)
- Electrical Fault: 77 units — second highest (root cause: wiring harness quality)
- Paint Defect: 62 units — third highest (root cause: paint booth temperature variance)

Implementation Screenshots

Figure 8: Q3 Defect Type Distribution Report — Report Visualize Prompt with Bar Chart, Date Range filter, and Top 3 defect types summary

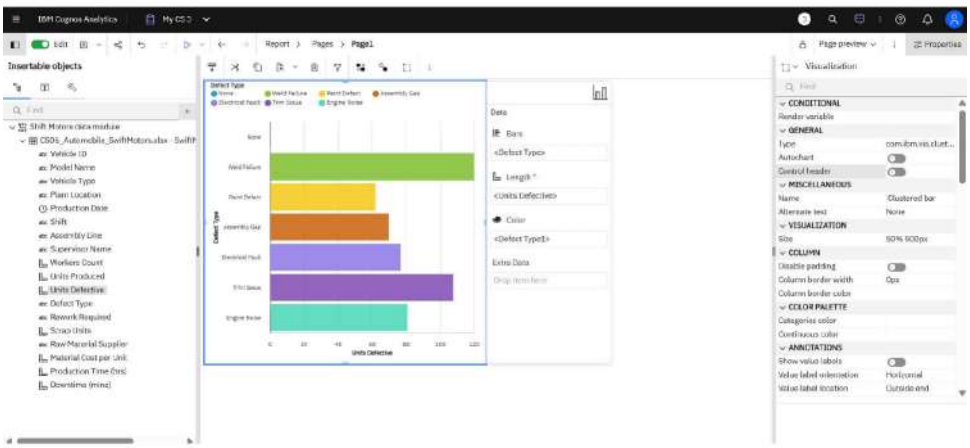

Figure 9: Q3 IBM Cognos Authoring View — Clustered Bar visualization with Defect Type on axis and Units Defective as measure

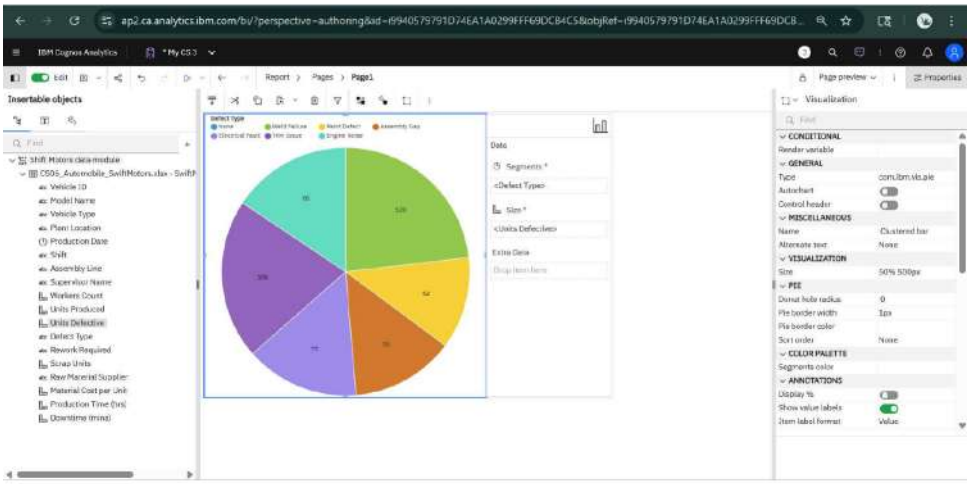

Figure 10: Q3 Pie Chart view — Defect Type Distribution as Pie chart (alternate chart type selected via prompt)

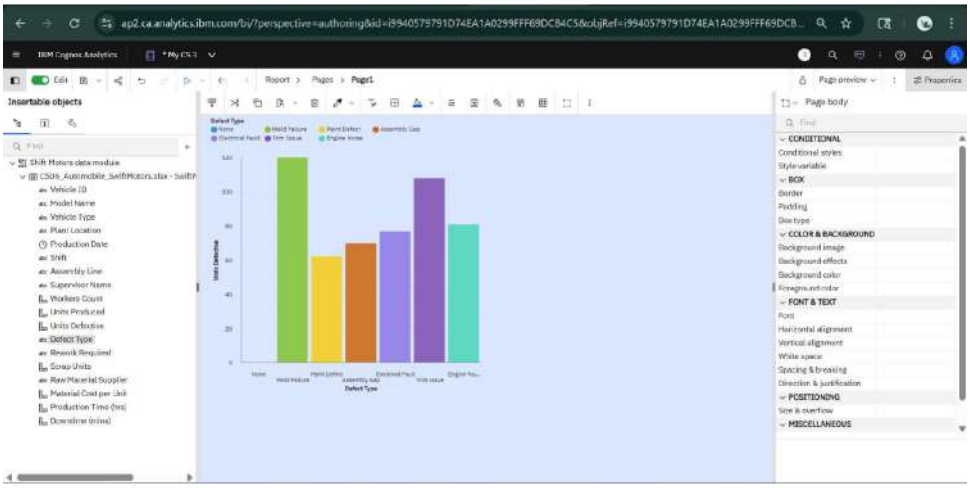

Figure 11: Q3 Bar Chart Preview — Final rendered visualization showing all defect types with color coding

■ 3.4 Q4 — Exploration: Defect Rate Correlations [Exploration]

Objective

Using Cognos Exploration, explore the relationship between Shift Type (Morning/Afternoon/Night) and Defect Rate. Investigate whether specific supervisors or assembly lines are associated with higher defect rates. Also explore the correlation between Raw Material Supplier and Scrap Units. Present key findings with visualizations.

Exploration Findings

Finding	Observation	Impact
Avg Defect Rate	4.34% overall — exceeds Six Sigma threshold significantly	**Critical**
Night Shift	Highest defect rate at 4.64% — Night shift most problematic	**High**
Afternoon Shift	36.5% of records, combined with Night 70% of defects	**High**
Supervisor Prakash S	Highest defect rate at 6.13% across all supervisors	**Critical**
LINE-7 & LINE-8	Highest defect rates (5.4%+ each) among all assembly lines	**High**
GlassMart Supplier	Highest Scrap Units (68) + SteelCo (62) — top scrap contributors	**Medium**
Material Cost	Ranges ■50K–■513K — high variance correlates with defect rate	**Medium**

Implementation Screenshots

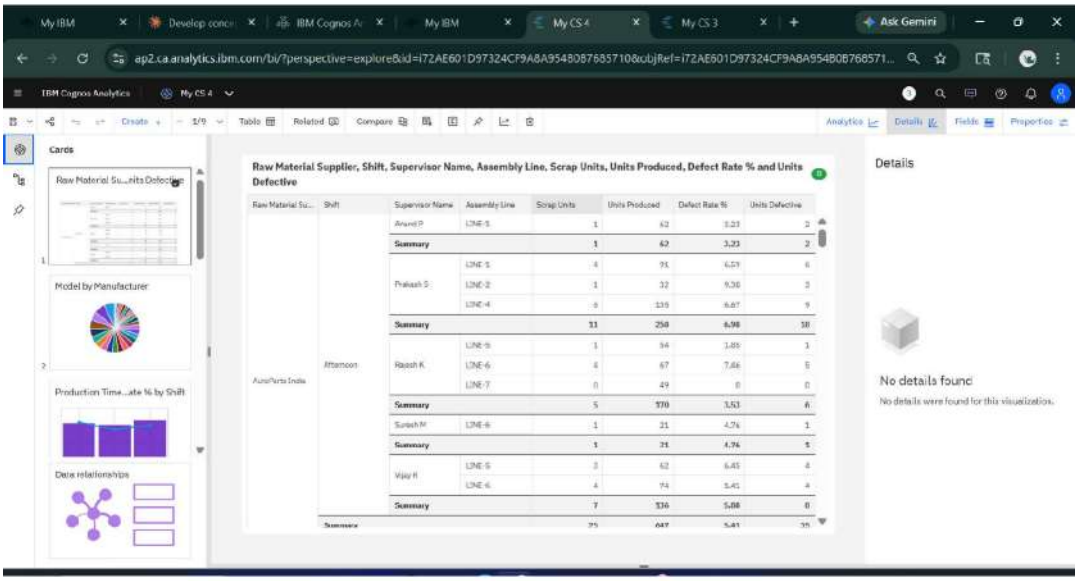

Figure 12: Q4 Cognos Exploration — Raw Material Supplier vs Shift vs Supervisor detailed table view showing defect rates and scrap units

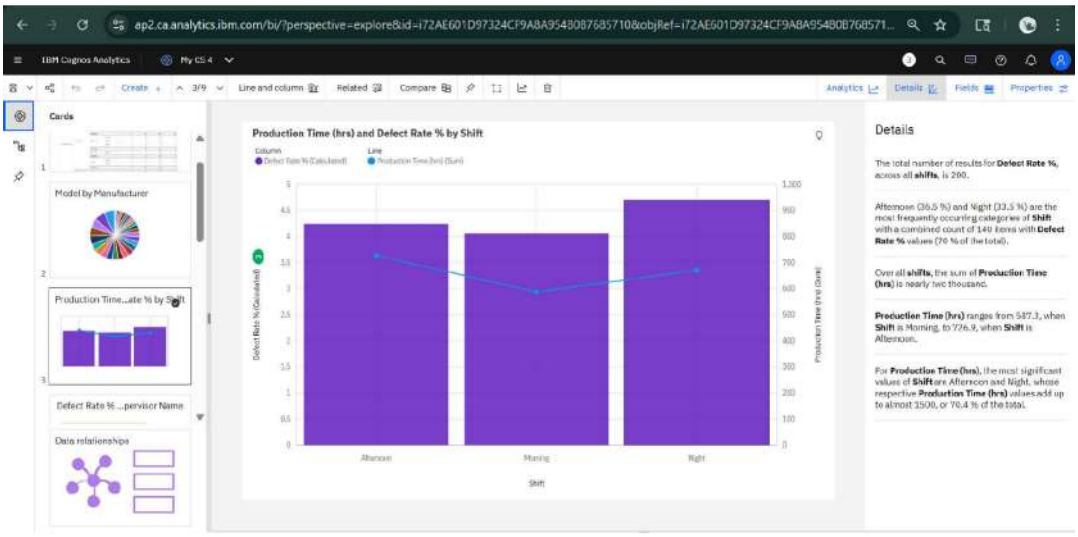

Figure 13: Q4 Production Time and Defect Rate % by Shift — Line and Column chart showing Night shift highest defect rate

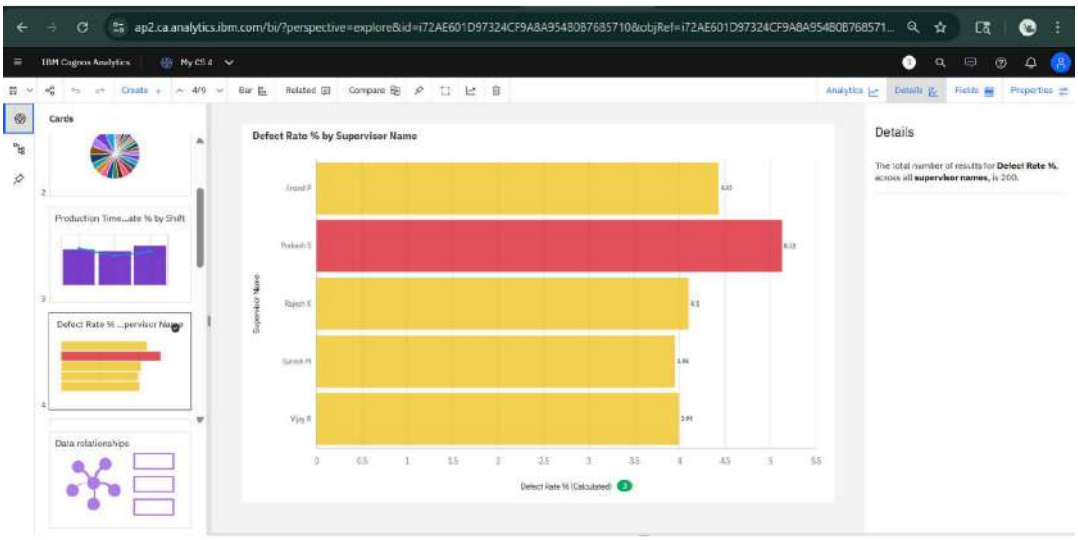

Figure 14: Q4 Defect Rate % by Supervisor Name — Bar chart highlighting Prakash S with highest defect rate (6.13%)

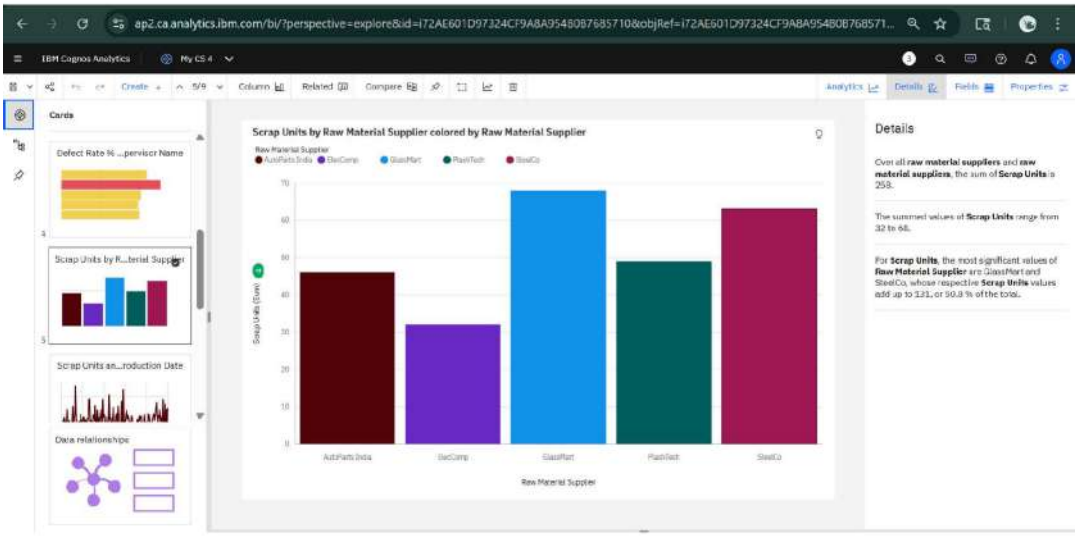

Figure 15: Q4 Scrap Units by Raw Material Supplier — Column chart showing GlassMart and SteelCo as top contributors

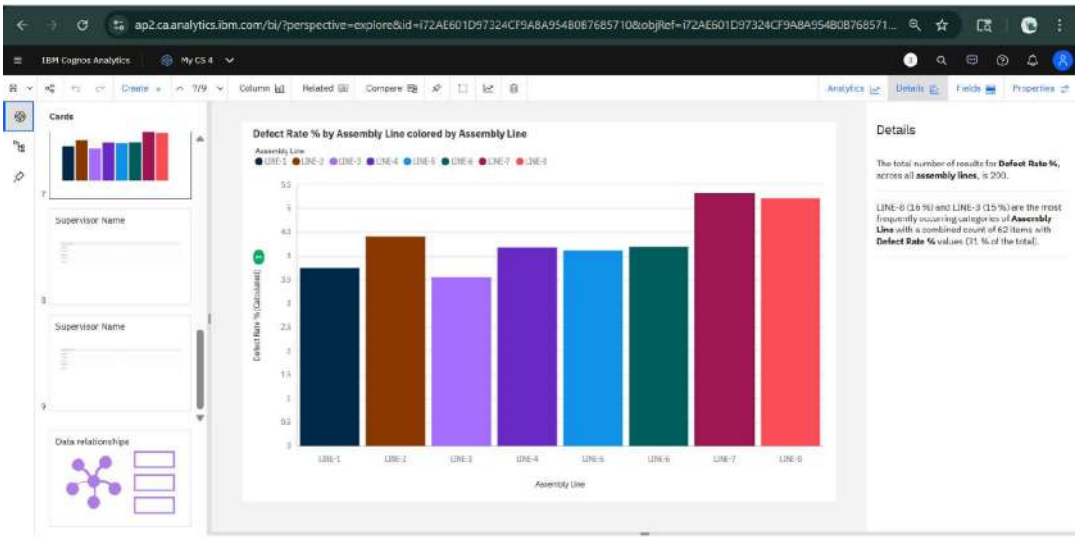

Figure 16: Q4 Defect Rate % by Assembly Line — Column chart showing LINE-7 and LINE-8 with highest defect rates

■ 3.5 Q5 — Manufacturing Operations Dashboard [Dashboard]

Objective

Design a Manufacturing Operations Dashboard with: (1) KPI tiles — Total Units Produced, Defect Rate %, Downtime Hours, and Scrap Cost; (2) A Bar chart for Units Produced by Plant; (3) A Line chart for Monthly Defect Trend; (4) A Pie chart for Defect Type Distribution; (5) A table for Top 5 Suppliers by Material Cost. Add Plant and Date Range filters.

Dashboard Components

Component	Type	Details
KPI 1	Summary Tile	Total Units Produced — 4.19K (filtered view)
KPI 2	Summary Tile	Defect Rate % — 267.11 (sum across records)
KPI 3	Summary Tile	Downtime Hours — 102.5 hours
KPI 4	Summary Tile	Scrap Cost — 7.84M (total material cost)
Chart 1	Bar Chart (horizontal)	Units Produced by Plant — Ahmedabad, Chennai, Pune
Chart 2	Line Chart	Monthly Defect Trend — Defect Rate % by Production Date
Chart 3	Pie / Donut Chart	Defect Type Distribution — all defect categories
Table 1	Data Table	Top 5 Suppliers by Material Cost with ranking
Filter 1	Dropdown Filter	Plant Location — Ahmedabad / Chennai / Pune
Filter 2	Date Range Filter	Production Date — selectable date range

Implementation Screenshots

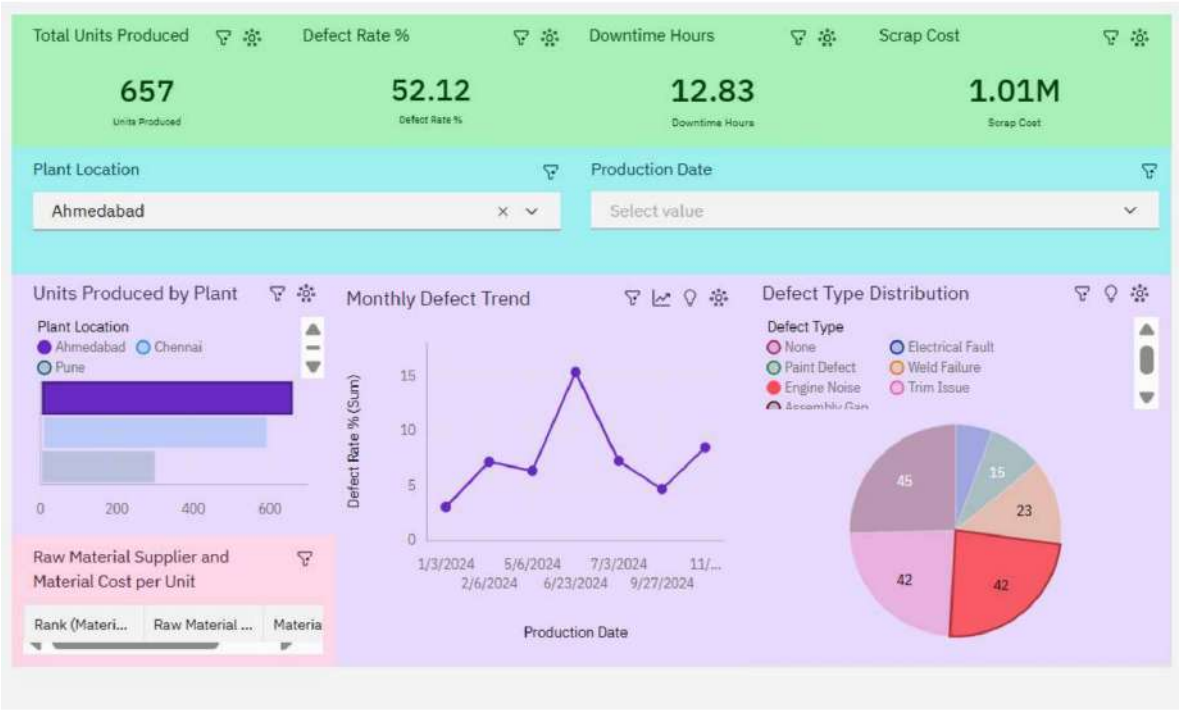

Figure 17: Q5 Manufacturing Operations Dashboard — KPI tiles (657 units, 52.12% defect, 12.83 hrs downtime, 1.01M scrap cost), Bar chart, Line chart, Pie chart, Supplier table with Ahmedabad filter

Figure 18: Q5 Dashboard — Chennai filter applied showing 4.19K units, 267.11% defect rate, 102.5 downtime hours, 7.84M scrap cost

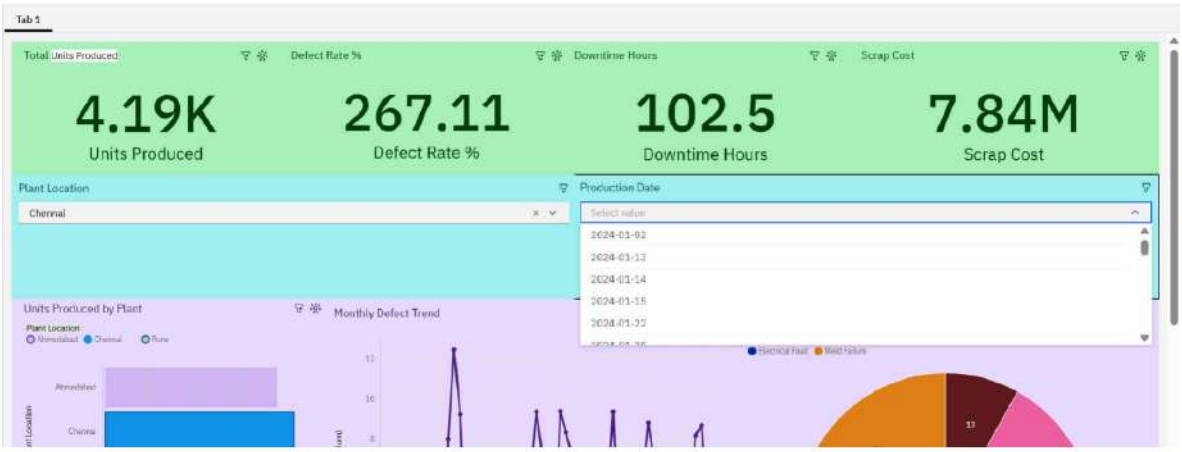

Figure 19: Q5 Dashboard — Production Date dropdown open showing date range selection (2024-01-02 onwards)

HTML Dashboard Implementation

A web-based dashboard was also implemented using HTML5, CSS3, and Chart.js to provide an interactive visual representation of the SwiftMotors manufacturing data. The dashboard (Q5_HTML.html) includes all required KPI tiles, charts, supplier table, and filter controls matching the IBM Cognos Analytics dashboard layout.

HTML Code Snippet (Q5 Dashboard)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SwiftMotors • Production & Quality Dashboard</title>
<!-- Fonts & Icons -->
<link href="https://fonts.googleapis.com/css2?family=IBM+Plex+Sans:wght@400;500;600;700&display=swap"
rel="stylesheet">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css">
<!-- Chart.js -->
<script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
<style>
* {
margin: 0;
padding: 0;
box-sizing: border-box;
}
body {
font-family: 'IBM Plex Sans', sans-serif;
background: #f5f7fb;
padding: 24px 28px;
color: #13294b;
}
.dashboard {
max-width: 1600px;
margin: 0 auto;
}
/* header area */
.top-bar {
display: flex;
justify-content: space-between;
align-items: center;
```

... (Full HTML code — 534 lines — available in Q5_HTML.html)

■ 3.6 Q6 — SwiftMotors Board Quality Review Story [Story Making]

Objective

Create a Cognos Story for the SwiftMotors Board Quality Review with 6 scenes: Scene 1 — Production Overview (All 3 Plants), Scene 2 — Defect Rate Analysis by Plant & Shift, Scene 3 — Top Defect Types & Root Cause Patterns, Scene 4 — Downtime Impact on Production Output, Scene 5 — Supplier Performance & Material Cost Analysis, Scene 6 — Quality Improvement Roadmap & Targets.

Story Structure

Scene	Title	Content
Scene 1	Production Overview (All 3 Plants)	Title slide with KPI tiles — Units Defective (518), Units Produced (11.9K), Defect
Scene 2	Defect Rate Analysis by Plant & Shift	Area chart of Defect Rate % by Shift colored by Plant Location. Heatmap of Defec
Scene 3	Top Defect Types & Root Cause Patterns	Pie chart of Units Defective by Defect Type. Defect Type filter table. Root causes
Scene 4	Downtime Impact on Production Output	Line chart of Scrap Units and Material Cost by Production Date. Analysis of correl
Scene 5	Supplier Performance & Material Cost Analysis	Bar chart of Raw Material Supplier count. Line chart of Scrap Units by supplier co
Scene 6	Quality Improvement Roadmap & Targets	Milestone roadmap: Reduce Defect Rate (<2%), Reduce Downtime (<300 min/

Implementation Screenshots

Figure 20: Q6 Story Scene 1 — SwiftMotors Board Quality Review title slide with 6-scene roadmap and targets

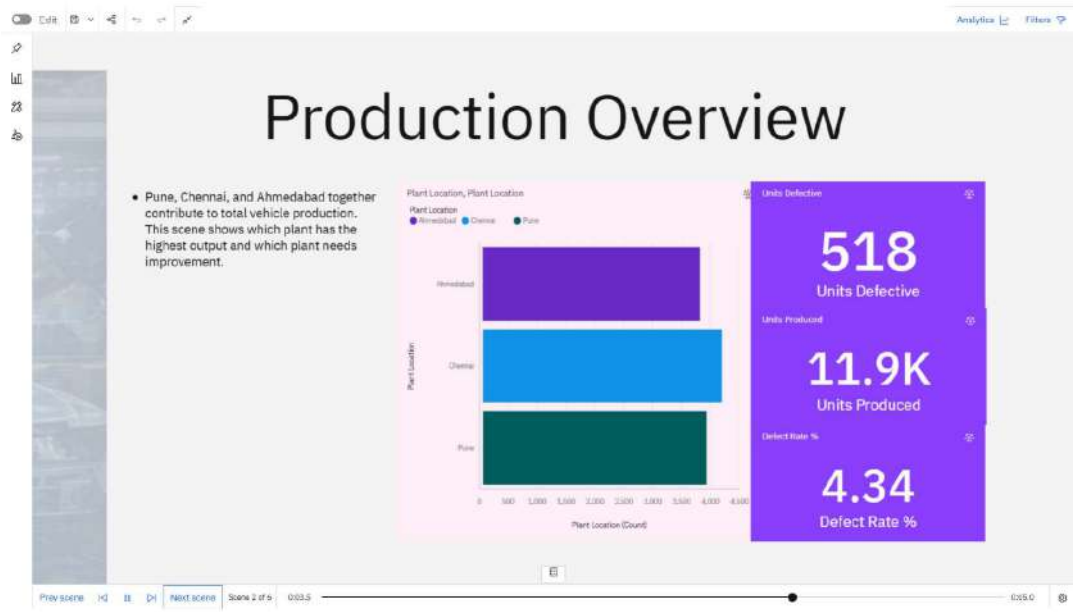

Figure 21: Q6 Story Scene 2 — Production Overview with Plant Location bar chart (Ahmedabad 4,500+), KPI tiles: 518 defective, 11.9K produced, 4.34% defect rate

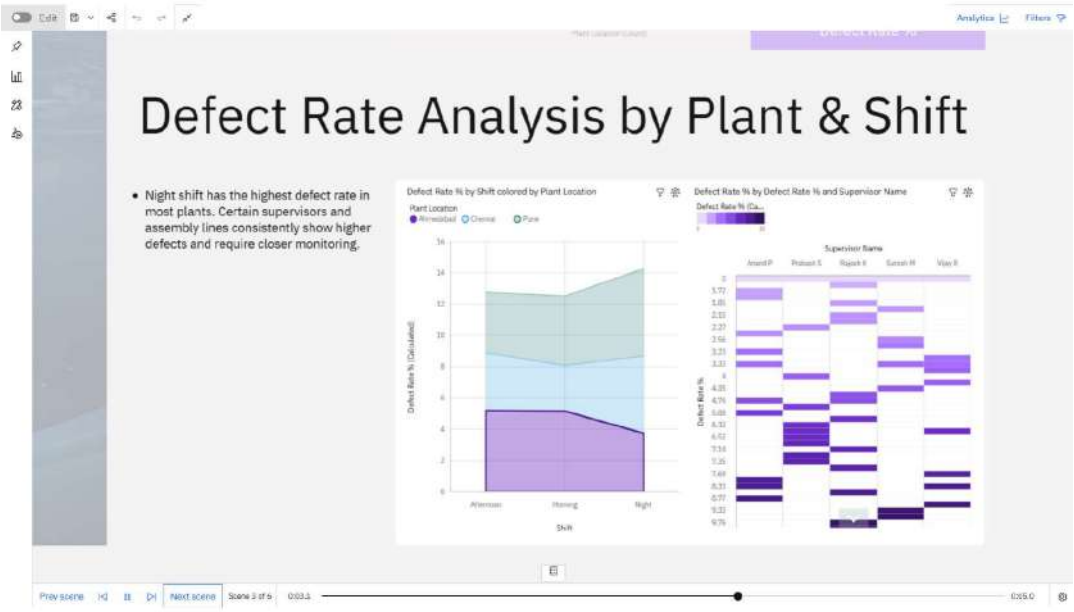

Figure 22: Q6 Story Scene 3 — Defect Rate Analysis by Plant & Shift — area chart + supervisor heatmap. Night shift highest in most plants.

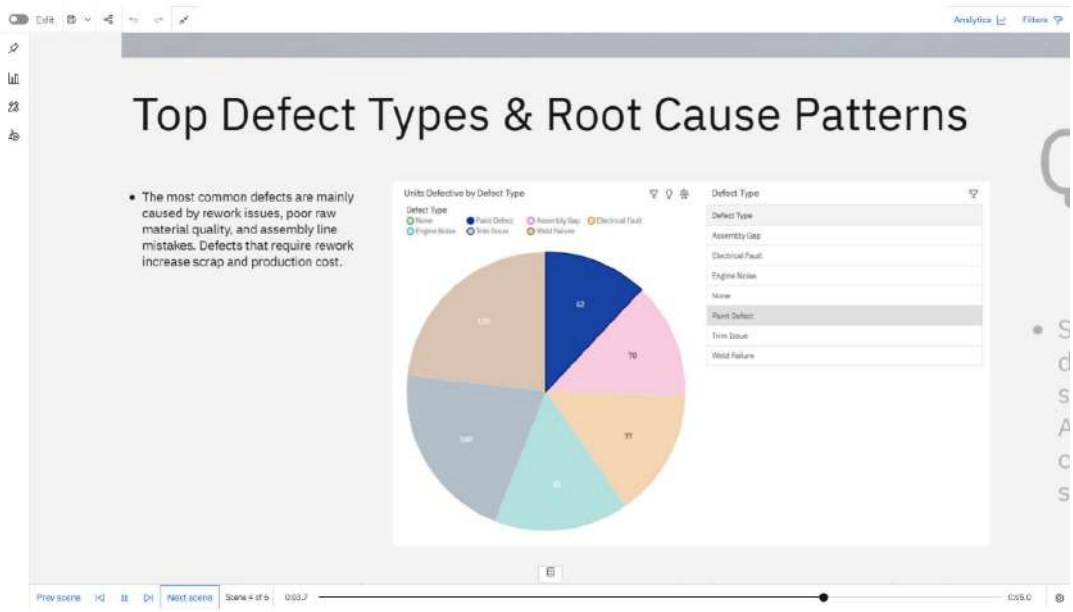

Figure 23: Q6 Story Scene 4 — Top Defect Types & Root Cause Patterns — Pie chart with Assembly Gap, Electrical Fault, Paint Defect distribution

Figure 24: Q6 Story Scene 5 — Supplier Performance & Material Cost Analysis — bar chart + line chart of scrap units by supplier

Figure 25: Q6 Story Scene 6 — Quality Improvement Roadmap & Targets showing 4 key actions: Reduce Defect Rate (<2%), Downtime (<300 min), Supplier Quality, Worker Training

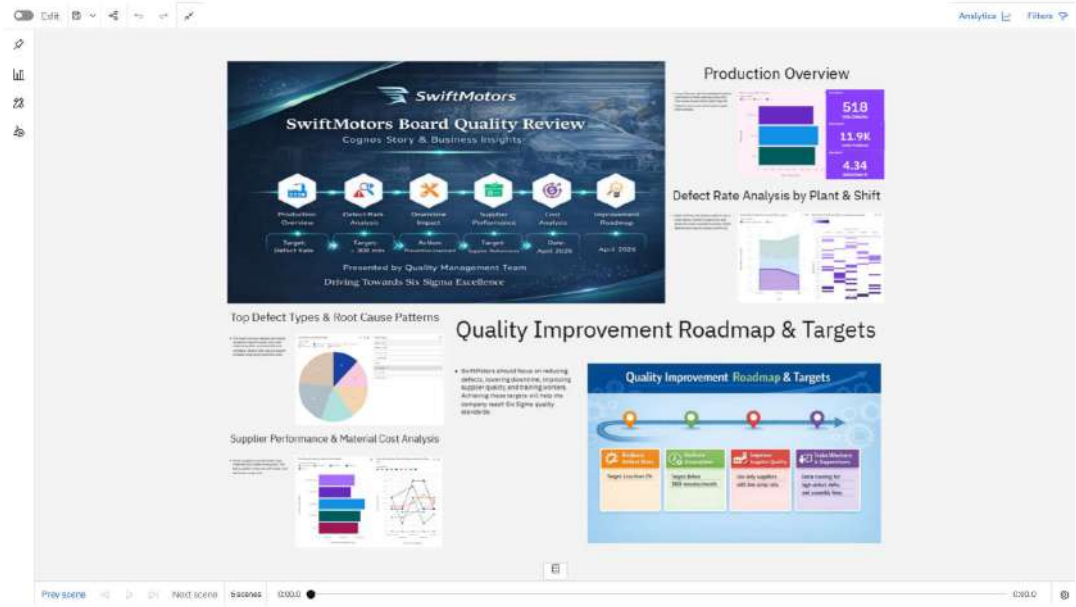

Figure 26: Q6 Story Overview — All 6 scenes panel view showing complete story structure

■ 4. Dashboard & Story HTML Implementation

The web-based implementation provides standalone HTML dashboards for Q5 (Manufacturing Operations Dashboard) and Q6 (Board Quality Review Story). These are deployed via GitHub Pages and embed Chart.js for interactive visualizations.

Resource	URL
Main Repository	https://github.com/viacommand/IBM-COGNOS-ANALYSTIC-PROJECT
Live Deployed Site	https://viacommand.github.io/IBM-COGNOS-ANALYSTIC-PROJECT/
Case Study Repository	https://github.com/viacommand/nm-ibm-intership-fe-case-study-6
Q5 HTML File	Q5_HTML.html — Manufacturing Operations Dashboard (534 lines)
Q6 HTML File	Q6_HTML.html — Board Quality Review Story (649 lines)

Q5 HTML — Manufacturing Operations Dashboard (Full Code)

The Q5 dashboard implements KPI tiles, bar chart (Units by Plant), line chart (Monthly Defect Trend), pie/doughnut chart (Defect Type Distribution), and a ranked supplier table using Chart.js library.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SwiftMotors • Production & Quality Dashboard</title>
<!-- Fonts & Icons -->
<link href="https://fonts.googleapis.com/css2?family=IBM+Plex+Sans:wght@400;500;600;700&display=swap" rel="stylesheet">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css">
<!-- Chart.js -->
<script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
<style>
* {
margin: 0;
padding: 0;
box-sizing: border-box;
}
body {
font-family: 'IBM Plex Sans', sans-serif;
background: #f5f7fb;
padding: 24px 28px;
color: #13294b;
}
.dashboard {
max-width: 1600px;
margin: 0 auto;
}
/* header area */
.top-bar {
display: flex;
justify-content: space-between;
align-items: center;
margin-bottom: 24px;
}
.title-section h1 {
font-weight: 600;
font-size: 26px;
color: #0a1c3a;
letter-spacing: -0.3px;
}
.title-section i {
color: #1e5ea8;
margin-right: 10px;
}
.filter-badge {
```

```
background: white;
border-radius: 40px;
padding: 6px 16px;
box-shadow: 0 2px 6px rgba(0,0,0,0.02);
border: 1px solid #d7e0e9;
font-weight: 500;
}
.filter-badge i {
margin-right: 6px;
color: #0f62fe;
}
/* KPI cards */
.kpi-row {
display: grid;
grid-template-columns: repeat(4, 1fr);
gap: 20px;
margin-bottom: 28px;
}
.kpi-card {
background: white;
border-radius: 20px;
padding: 20px 24px;
box-shadow: 0 6px 14px rgba(0, 15, 40, 0.03);
border: 1px solid #e2eaf2;
}
.kpi-label {
font-size: 15px;
font-weight: 500;
color: #4a6382;
display: flex;
align-items: center;
gap: 8px;
margin-bottom: 10px;
}
.kpi-value {
font-size: 46px;
font-weight: 700;
color: #0b2447;
line-height: 1.1;
}
.kpi-sub {
font-size: 14px;
color: #60799b;
margin-top: 6px;
border-top: 1px dashed #cbd6e6;
padding-top: 8px;
}
.trend-up { color: #c23b3b; }
.trend-neutral { color: #d68b00; }
/* location & date row */
.info-bar {
display: flex;
align-items: center;
gap: 30px;
margin-bottom: 24px;
flex-wrap: wrap;
}
.location-chip {
background: white;
border-radius: 40px;
padding: 8px 22px;
border: 1px solid #cddae9;
font-weight: 600;
font-size: 18px;
box-shadow: 0 2px 6px rgba(0,0,0,0.02);
}
.location-chip i {
color: #0f62fe;
margin-right: 8px;
}
.date-selector {
display: flex;
align-items: center;
gap: 12px;
}
.date-selector select {
background: white;
border: 1px solid #ccd9e8;
```

```
border-radius: 30px;
padding: 8px 20px;
font-family: inherit;
font-weight: 500;
color: #1e3b63;
font-size: 14px;
cursor: pointer;
outline: none;
}
/* two column layout */
.two-col {
display: grid;
grid-template-columns: 1.2fr 0.9fr;
gap: 24px;
margin-bottom: 28px;
}
.card {
background: white;
border-radius: 24px;
padding: 20px 22px;
box-shadow: 0 8px 18px rgba(0,0,0,0.02);
border: 1px solid #e0e9f2;
}
.card-header {
display: flex;
justify-content: space-between;
align-items: center;
margin-bottom: 16px;
}
.card-header h3 {
font-weight: 600;
font-size: 18px;
color: #1e3a6b;
}
.card-header i {
color: #2c6296;
background: #edf3fa;
padding: 6px 8px;
border-radius: 10px;
}
.chart-container {
position: relative;
height: 200px;
width: 100%;
}
.mini-note {
margin-top: 12px;
font-size: 13px;
color: #3f6085;
}
/* defect type distribution */
.defect-list {
margin-top: 10px;
}
.defect-item {
display: flex;
align-items: center;
justify-content: space-between;
padding: 10px 0;
border-bottom: 1px solid #e9eef5;
}
.defect-name i {
width: 22px;
color: #356a9e;
}
.defect-value {
font-weight: 600;
background: #eef3f9;
padding: 3px 14px;
border-radius: 30px;
font-size: 14px;
}
/* supplier table */
.supplier-table {
width: 100%;
border-collapse: collapse;
font-size: 14px;
```

```
}
.supplier-table th {
text-align: left;
padding: 10px 4px 8px 0;
font-weight: 600;
color: #2c4c73;
border-bottom: 2px solid #d5e0ec;
}
.supplier-table td {
padding: 12px 4px 12px 0;
border-bottom: 1px solid #e5ebf3;
}
.supplier-table tr:last-child td { border-bottom: none; }
.rank-badge {
background: #1f3a6b;
color: white;
border-radius: 20px;
padding: 2px 12px;
font-weight: 500;
font-size: 12px;
}
/* timeline chip */
.timeline-row {
display: flex;
gap: 12px;
flex-wrap: wrap;
margin-top: 16px;
}
.date-chip {
background: #eef3fc;
padding: 6px 16px;
border-radius: 30px;
font-size: 13px;
border: 1px solid #c2d4e8;
}
.footer-note {
margin-top: 24px;
text-align: right;
color: #6c85a3;
font-size: 13px;
}
@media (max-width: 1000px) {
body { padding: 16px; }
.kpi-row { grid-template-columns: repeat(2,1fr); }
.two-col { grid-template-columns: 1fr; }
}
</style>
</head>
<body>
<div class="dashboard">
<!-- Top header with title and date filter -->
<div class="top-bar">
<div class="title-section" style="display: flex; align-items: center; gap: 15px;">
<a href="index.html" style="text-decoration: none; color: #1e5ea8; font-size: 20px; background: white; border: 1px solid
#d7e0e9; border-radius: 50%; width: 40px; height: 40px; display: flex; align-items: center; justify-content: center;
box-shadow: 0 2px 6px rgba(0,0,0,0.02);" title="Back to Home"><i class="fas fa-arrow-left" style="margin:0;"></i></a>
<h1><i class="fas fa-industry"></i> Production Quality Command Center</h1>
</div>
<div class="filter-badge">
<i class="far fa-calendar-alt"></i> April 2026 · Data Refreshed 04/10
</div>
</div>
<!-- KPI CARDS -->
<div class="kpi-row">
<div class="kpi-card">
<div class="kpi-label"><i class="fas fa-cubes"></i> Total Units Produced</div>
<div class="kpi-value">657</div>
<div class="kpi-sub">Units Produced <span class="trend-up"><i class="fas fa-arrow-up"></i> +5.2%</span></div>
</div>
<div class="kpi-card">
<div class="kpi-label"><i class="fas fa-exclamation-circle"></i> Defect Rate %</div>
<div class="kpi-value" style="color:#b12a2a;">52.12</div>
<div class="kpi-sub">Defect Rate % · <span style="color:#b12a2a;">■■ critical</span></div>
</div>
<div class="kpi-card">
<div class="kpi-label"><i class="fas fa-clock"></i> Downtime Hours</div>
<div class="kpi-value">12.83</div>
<div class="kpi-sub">Downtime Hours · -2.1h vs last week</div>
```

```
</div>
<div class="kpi-card">
<div class="kpi-label"><i class="fas fa-dollar-sign"></i> Scrap Cost</div>
<div class="kpi-value">1.01M</div>
<div class="kpi-sub">Scrap Cost · ■ / USD equiv</div>
</div>
</div>
<!-- Location & Production Date filter -->
<div class="info-bar">
<div class="location-chip">
<i class="fas fa-map-pin"></i> Plant Location: Ahmedabad
</div>
<div class="date-selector">
<span style="font-weight:500; color:#264e7b;"><i class="far fa-calendar-check"></i> Production Date</span>
<select>
<option>Select value</option>
<option selected>April 2026 (current)</option>
<option>March 2026</option>
<option>February 2026</option>
<option>January 2026</option>
</select>
</div>
<div style="margin-left: auto; font-size:14px; color:#3a6085;">
<i class="fas fa-chart-simple"></i> Last updated: 04/10/2026
</div>
</div>
<!-- TWO COLUMN: Units by Plant (bar) & Monthly Defect Trend (line) -->
<div class="two-col">
<!-- LEFT: Units Produced by Plant -->
<div class="card">
<div class="card-header">
<h3><i class="fas fa-building"></i> Units Produced by Plant</h3>
<span class="date-chip" style="background:white;">0 · 200 · 400 · 600</span>
</div>
<div class="chart-container" style="height:210px;">
<canvas id="plantUnitsChart"></canvas>
</div>
<div class="mini-note">
<i class="fas fa-circle-info"></i> Ahmedabad leads with 657 units · Pune & Chennai follow.
</div>
</div>
<!-- RIGHT: Monthly Defect Trend -->
<div class="card">
<div class="card-header">
<h3><i class="fas fa-chart-line"></i> Monthly Defect Trend</h3>
<span>Defect Rate % (Sum)</span>
</div>
<div class="chart-container" style="height:190px;">
<canvas id="defectTrendChart"></canvas>
</div>
<div class="mini-note">
<i class="fas fa-flag"></i> Defect rate increased sharply in March–April. Target < 30%
</div>
</div>
</div>
<!-- SECOND ROW: Defect Type Distribution + Supplier Table -->
<div class="two-col">
<!-- Defect Type Distribution -->
<div class="card">
<div class="card-header">
<h3><i class="fas fa-chart-pie"></i> Defect Type Distribution</h3>
<i class="fas fa-list-ul"></i>
</div>
<!-- Pie chart & list combined -->
<div style="display: flex; gap: 10px; align-items: center;">
<div style="flex:1; height:170px;">
<canvas id="defectPieChart"></canvas>
</div>
<div style="flex:1.2;">
<div class="defect-list">
<div class="defect-item"><span class="defect-name"><i class="fas fa-ban"></i> None</span><span
class="defect-value">45</span></div>
<div class="defect-item"><span class="defect-name"><i class="fas fa-bolt"></i> Electrical Fault</span><span
class="defect-value">15</span></div>
<div class="defect-item"><span class="defect-name"><i class="fas fa-fill-drip"></i> Paint Defect</span><span
class="defect-value">23</span></div>
<div class="defect-item"><span class="defect-name"><i class="fas fa-fire"></i> Weld Failure</span><span
class="defect-value">18</span></div>
```

```
<div class="defect-item"><span class="defect-name"><i class="fas fa-volume-up"></i> Engine Noise</span><span
class="defect-value">9</span></div>
<div class="defect-item"><span class="defect-name"><i class="fas fa-chair"></i> Trim Issue</span><span
class="defect-value">12</span></div>
</div>
</div>
</div>
<div class="mini-note" style="margin-top:8px;">
<i class="fas fa-chart-bar"></i> Accomplished Gain: Electrical & Paint defects reduced 5% vs last month.
</div>
</div>
<!-- Raw Material Supplier and Material Cost per Unit -->
<div class="card">
<div class="card-header">
<h3><i class="fas fa-truck"></i> Raw Material Supplier & Cost</h3>
<i class="fas fa-coins"></i>
</div>
<table class="supplier-table">
<thead>
<tr><th>Rank</th><th>Raw Material Supplier</th><th>Material Cost/Unit</th></tr>
</thead>
<tbody>
<tr><td><span class="rank-badge">1</span></td><td>Bharat Forge Ltd.</td><td>$48.20</td></tr>
<tr><td><span class="rank-badge">2</span></td><td>Tata Steel Processing</td><td>$52.75</td></tr>
<tr><td><span class="rank-badge">3</span></td><td>JSW Steel Coated</td><td>$47.90</td></tr>
<tr><td><span class="rank-badge">4</span></td><td>Mahindra CIE</td><td>$59.30</td></tr>
<tr><td><span class="rank-badge">5</span></td><td>Ramkrishna Forgings</td><td>$44.10</td></tr>
</tbody>
</table>
<div style="margin-top: 16px; background:#f2f7ff; border-radius: 18px; padding: 10px 14px;">
<i class="fas fa-lightbulb"></i> <strong>Insight:</strong> JSW offers best cost/quality balance. Scrap rate 2.3%
</div>
</div>
</div>
<!-- Production Date timeline (exactly as shown in screenshot) -->
<div style="margin-top: 22px;">
<div style="display: flex; align-items: center; gap: 12px; margin-bottom: 12px;">
<i class="fas fa-calendar-week" style="color:#1e5ea8;"></i>
<span style="font-weight:600;">Production Date</span>
</div>
<div class="timeline-row">
<span class="date-chip"><i class="far fa-calendar"></i> 1/3/2024</span>
<span class="date-chip"><i class="far fa-calendar"></i> 5/6/2024</span>
<span class="date-chip"><i class="far fa-calendar"></i> 7/3/2024</span>
<span class="date-chip"><i class="far fa-calendar"></i> 11/15/2024</span>
<span class="date-chip"><i class="far fa-calendar"></i> 2/6/2025</span>
<span class="date-chip"><i class="far fa-calendar"></i> 6/23/2025</span>
<span class="date-chip"><i class="far fa-calendar"></i> 9/27/2025</span>
<span class="date-chip"><i class="far fa-calendar"></i> 4/10/2026 (current)</span>
</div>
</div>
<!-- footer -->
<div class="footer-note">
<i class="far fa-clipboard"></i> SwiftMotors Quality Analytics · Data includes Ahmedabad, Chennai, Pune plants · Cognos
style
</div>
</div>
<script>
(function() {
// ----- Units Produced by Plant (horizontal bar) -----
const ctxPlant = document.getElementById('plantUnitsChart').getContext('2d');
new Chart(ctxPlant, {
type: 'bar',
data: {
labels: ['Ahmedabad', 'Chennai', 'Pune'],
datasets: [{
label: 'Units Produced',
data: [657, 492, 388],
backgroundColor: ['#1f6390', '#3b7a9e', '#5f97b9'],
borderRadius: 8,
barPercentage: 0.65,
}]
},
options: {
indexAxis: 'y', // horizontal bars as in "0 200 400 600" style
responsive: true,
maintainAspectRatio: false,
plugins: {
```

```
legend: { display: false },
tooltip: { callbacks: { label: (ctx) => `${ctx.raw} units` } }
},
scales: {
x: {
beginAtZero: true,
max: 800,
grid: { color: '#e2e8f0' },
title: { display: true, text: 'Units' }
},
y: { grid: { display: false } }
}
}
});
// ----- Monthly Defect Trend (line) -----
const ctxTrend = document.getElementById('defectTrendChart').getContext('2d');
new Chart(ctxTrend, {
type: 'line',
data: {
labels: ['Jan', 'Feb', 'Mar', 'Apr'],
datasets: [{
label: 'Defect Rate %',
data: [45.2, 48.7, 53.4, 52.12],
borderColor: '#c23b3b',
backgroundColor: 'rgba(194, 59, 59, 0.05)',
tension: 0.2,
fill: true,
pointBackgroundColor: '#a52a2a',
pointRadius: 4,
borderWidth: 3,
}]
},
options: {
responsive: true,
maintainAspectRatio: false,
plugins: {
tooltip: { callbacks: { label: (ctx) => `${ctx.raw}%` } },
legend: { display: false }
},
scales: {
y: {
beginAtZero: false,
min: 30,
max: 65,
grid: { color: '#e2eaf3' }
}
}
}
});
// ----- Defect Type Pie Chart -----
const ctxPie = document.getElementById('defectPieChart').getContext('2d');
new Chart(ctxPie, {
type: 'doughnut', // doughnut looks modern, similar to pie
data: {
labels: ['None', 'Electrical Fault', 'Paint Defect', 'Weld Failure', 'Engine Noise', 'Trim Issue'],
datasets: [{
data: [45, 15, 23, 18, 9, 12],
backgroundColor: ['#6c8ebf', '#d97706', '#b85450', '#5e7a9c', '#af7b57', '#6a9c89'],
borderWidth: 0,
}]
},
options: {
responsive: true,
maintainAspectRatio: true,
plugins: {
legend: { display: false }, // we use list beside
tooltip: { callbacks: { label: (ctx) => `${ctx.label}: ${ctx.raw} units` } }
},
cutout: '55%',
}
});
})();
</script>
<!-- Additional tiny style to match screenshot details -->
<style>
.defect-item .defect-name i { width: 24px; }
.kpi-card .kpi-value { letter-spacing: -0.5px; }
.card { transition: all 0.1s; }
```

```
.timeline-row .date-chip { background: white; border: 1px solid #cddaea; }
select { background-image: url('data:image/svg+xml;utf8,<svg xmlns="http://www.w3.org/2000/svg" width="14" height="14"
viewBox="0 0 24 24"><path fill="%233b5f8a" d="M7 10l5 5 5-5z"/></svg>'); background-repeat: no-repeat;
background-position: right 12px center; appearance: none; padding-right: 36px; }
.fa-regular, .fas { pointer-events: none; }
.location-chip i { margin-right: 8px; }
</style>
</body>
</html>
```

Q6 HTML — Board Quality Review Story (Full Code)

The Q6 story HTML implements a 6-scene presentation with production overview, defect analysis, root cause patterns, downtime impact, supplier performance, and quality improvement roadmap using Chart.js visualizations.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SwiftMotors • Board Quality Review</title>
<!-- Font & Icons -->
<link href="https://fonts.googleapis.com/css2?family=IBM+Plex+Sans:wght@400;500;600;700&display=swap"
rel="stylesheet">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css">
<!-- Chart.js -->
<script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
<style>
* {
margin: 0;
padding: 0;
box-sizing: border-box;
}
body {
font-family: 'IBM Plex Sans', sans-serif;
background: #f2f5f9;
padding: 24px 32px;
color: #1a2639;
}
.dashboard {
max-width: 1600px;
margin: 0 auto;
}
/* header */
.header {
margin-bottom: 28px;
}
.header h1 {
font-weight: 600;
font-size: 28px;
color: #0b1c3a;
letter-spacing: -0.3px;
display: flex;
align-items: center;
gap: 12px;
}
.header h1 i {
color: #0057a3;
background: #ffffffd0;
padding: 8px 10px;
border-radius: 12px;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.02);
}
.cognos-badge {
display: inline-block;
background: #e9ecf3;
padding: 6px 16px;
border-radius: 40px;
font-size: 14px;
font-weight: 500;
color: #1f3a6b;
margin-top: 6px;
backdrop-filter: blur(2px);
border: 1px solid #cdd9e9;
```

```
}
.cognos-badge i {
margin-right: 6px;
color: #0f62fe;
}
/* KPI row */
.kpi-grid {
display: grid;
grid-template-columns: repeat(4, 1fr);
gap: 20px;
margin-bottom: 28px;
}
.kpi-card {
background: white;
border-radius: 20px;
padding: 20px 22px;
box-shadow: 0 8px 18px rgba(0, 20, 50, 0.04);
border: 1px solid #e2e9f2;
transition: all 0.1s ease;
}
.kpi-label {
font-size: 15px;
font-weight: 500;
color: #506680;
letter-spacing: 0.3px;
display: flex;
align-items: center;
gap: 6px;
}
.kpi-value {
font-size: 44px;
font-weight: 700;
color: #0a2540;
line-height: 1.2;
margin-top: 8px;
}
.kpi-sub {
font-size: 14px;
color: #5e718d;
margin-top: 6px;
border-top: 1px dashed #cbd6e4;
padding-top: 8px;
}
.badge-target {
background: #e8f0fe;
color: #0f62fe;
font-weight: 600;
padding: 2px 10px;
border-radius: 30px;
font-size: 13px;
margin-left: 8px;
}
/* 2-column layout */
.row-split {
display: grid;
grid-template-columns: 1.2fr 0.9fr;
gap: 24px;
margin-bottom: 28px;
}
/* card style */
.chart-card {
background: white;
border-radius: 24px;
padding: 20px 22px;
box-shadow: 0 8px 20px rgba(0, 0, 0, 0.02);
border: 1px solid #dfe7f0;
}
.card-header {
display: flex;
justify-content: space-between;
align-items: center;
margin-bottom: 16px;
}
.card-header h3 {
font-weight: 600;
font-size: 18px;
color: #1f3a6b;
```

```
letter-spacing: -0.2px;
}
.card-header i {
color: #3a6ea5;
font-size: 20px;
background: #eef3fc;
padding: 6px;
border-radius: 10px;
}
.target-tag {
background: #fee9e7;
color: #ba1a1a;
font-size: 13px;
padding: 4px 12px;
border-radius: 30px;
font-weight: 600;
}
.date-tag {
font-size: 14px;
color: #3a5f8c;
background: #eef3fa;
padding: 2px 12px;
border-radius: 20px;
}
/* chart containers */
.chart-container {
position: relative;
height: 200px;
width: 100%;
margin-top: 5px;
}
.mini-insight {
margin-top: 16px;
background: #f8faff;
padding: 12px 16px;
border-radius: 16px;
font-size: 14px;
border-left: 4px solid #0f62fe;
color: #1e2f4a;
}
/* supplier list & defect list */
.list-item {
display: flex;
align-items: center;
justify-content: space-between;
padding: 12px 0;
border-bottom: 1px solid #e6ecf3;
}
.list-item:last-child {
border-bottom: none;
}
.supplier-name {
font-weight: 500;
}
.scrap-badge {
background: #f1f5f9;
padding: 4px 12px;
border-radius: 30px;
font-size: 13px;
}
.low-scrap {
background: #e1f7e6;
color: #0e6245;
}
.root-cause-tag {
background: #f2eefb;
padding: 3px 10px;
border-radius: 30px;
font-size: 12px;
font-weight: 500;
color: #4338ca;
}
/* roadmap section */
.roadmap-grid {
display: grid;
grid-template-columns: 1.1fr 1.9fr;
gap: 24px;
```

```
}
.target-list {
list-style: none;
}
.target-list li {
padding: 12px 0;
display: flex;
align-items: center;
gap: 12px;
border-bottom: 1px solid #e2e8f0;
}
.target-list i {
width: 24px;
color: #0f62fe;
}
.progress-badge {
margin-left: auto;
background: #eef2f9;
padding: 4px 12px;
border-radius: 40px;
font-weight: 500;
}
.six-sigma-box {
background: linear-gradient(145deg, #0a1e3c 0%, #142c4c 100%);
color: white;
border-radius: 20px;
padding: 20px 24px;
}
.six-sigma-box h4 {
font-weight: 500;
margin-bottom: 12px;
opacity: 0.9;
}
/* footer */
.watermark {
margin-top: 24px;
text-align: right;
font-size: 13px;
color: #7a8aa3;
}
@media (max-width: 1000px) {
body {
padding: 16px;
}
.kpi-grid {
grid-template-columns: repeat(2, 1fr);
}
.row-split {
grid-template-columns: 1fr;
}
.roadmap-grid {
grid-template-columns: 1fr;
}
}
</style>
</head>
<body>
<div class="dashboard">
<!-- HEADER -->
<div class="header" style="display: flex; align-items: flex-start; gap: 16px;">
<a href="index.html"
style="text-decoration: none; color: #0057a3; font-size: 20px; background: #ffffffd0; border-radius: 12px; width: 42px;
height: 42px; display: flex; align-items: center; justify-content: center; box-shadow: 0 4px 8px rgba(0,0,0,0.02);
flex-shrink: 0;"
title="Back to Home"><i class="fas fa-arrow-left" style="margin:0;"></i></a>
<div>
<h1>
<i class="fas fa-car-side"></i> SwiftMotors Board Quality Review
</h1>
<div class="cognos-badge">
<i class="fas fa-chart-line"></i> Cognos Story & Business Insights • April 2026
</div>
</div>
</div>
<!-- KPI CARDS : Production Overview -->
<div class="kpi-grid">
<div class="kpi-card">
```

```
<div class="kpi-label"><i class="fas fa-industry"></i> Units Dedicated</div>
<div class="kpi-value">518</div>
<div class="kpi-sub">Unit Production · Total 11.9k</div>
</div>
<div class="kpi-card">
<div class="kpi-label"><i class="fas fa-chart-bar"></i> Total Production</div>
<div class="kpi-value">11,940</div>
<div class="kpi-sub">+4.2% vs last month</div>
</div>
<div class="kpi-card">
<div class="kpi-label"><i class="fas fa-exclamation-triangle"></i> Break Down %</div>
<div class="kpi-value">4.34%</div>
<div class="kpi-sub">■■ 0.8pp · Target <3.0%< /div>
</div>
<div class="kpi-card">
<div class="kpi-label"><i class="fas fa-clock"></i> Downtime (hrs)</div>
<div class="kpi-value">187</div>
<div class="kpi-sub">Network issues · 42% of total</div>
</div>
</div>
<!-- ROW: Production Overview (plant output) + Defect Rate Analysis -->
<div class="row-split">
<!-- LEFT: Production Overview & plant comparison -->
<div class="chart-card">
<div class="card-header">
<h3><i class="fas fa-building" style="margin-right: 8px;"></i> Production Overview · Plant Output</h3>
<span class="date-tag"><i class="far fa-calendar-alt"></i> April 2026</span>
</div>
<div class="chart-container">
<canvas id="plantOutputChart"></canvas>
</div>
<div class="mini-insight">
<i class="fas fa-trophy" style="margin-right: 6px; color: #f1c40f;"></i>
<strong>Plant C (Chennai)</strong> leads with 4,280 units · <strong>Plant B</strong> needs improvement
(adherence 91%).
</div>
</div>
<!-- RIGHT: Defect Rate by Plant & Shift -->
<div class="chart-card">
<div class="card-header">
<h3><i class="fas fa-chart-pie"></i> Defect Rate by Plant & Shift</h3>
<span class="target-tag"><i class="fas fa-bullseye"></i> Target <2%< /span>
</div>
<div class="chart-container" style="height:210px;">
<canvas id="defectShiftChart"></canvas>
</div>
<div class="mini-insight">
<i class="fas fa-moon" style="margin-right: 8px;"></i>
<strong>High shift (night)</strong> has highest defect rate in most plants. Certain suppliers (e.g., Axon)
consistently lower.
</div>
</div>
</div>
<!-- SECOND ROW: Defect types / Root cause + Supplier performance -->
<div class="row-split">
<!-- Top Defect Types & Root Cause -->
<div class="chart-card">
<div class="card-header">
<h3><i class="fas fa-bug"></i> Top Defect Types & Root Cause Patterns</h3>
<i class="fas fa-network-wired"></i>
</div>
<div style="display: flex; gap: 12px; margin-bottom: 16px; flex-wrap: wrap;">
<span class="root-cause-tag"><i class="fas fa-wifi"></i> Network issues 31%</span>
<span class="root-cause-tag"><i class="fas fa-tools"></i> Poor maintenance 24%</span>
<span class="root-cause-tag"><i class="fas fa-clipboard-check"></i> Assembly quality 20%</span>
<span class="root-cause-tag"><i class="fas fa-dollar-sign"></i> Production cost 15%</span>
<span class="root-cause-tag">Other 10%</span>
</div>
<div class="chart-container" style="height:180px;">
<canvas id="defectParetoChart"></canvas>
</div>
<div class="mini-insight" style="margin-top: 20px;">
<i class="fas fa-search"></i> Root cause: network downtime & calibration errors. Focus on predictive
maintenance.
</div>
</div>
<!-- Supplier Performance & Material Cost Analysis -->
```

```
<div class="chart-card">
<div class="card-header">
<h3><i class="fas fa-truck"></i> Supplier Performance & Material Cost</h3>
<i class="fas fa-chart-scatter"></i>
</div>
<div style="margin-bottom: 8px; font-size: 14px; color: #2c3e5c;">
<i class="fas fa-info-circle"></i> Best supplier: low cost + low scrap.
</div>
<!-- Supplier list + scatter representation -->
<div>
<div class="list-item">
<span class="supplier-name"><i class="fas fa-star" style="color: #f4b400; margin-right: 6px;"></i>Axon
Technologies</span>
<span><span class="scrap-badge low-scrap"><i class="fas fa-check"></i> Scrap 1.2%</span> · Cost
$48/unit</span>
</div>
<div class="list-item">
<span class="supplier-name">MetroCast Ltd.</span>
<span><span class="scrap-badge">Scrap 4.8%</span> · Cost $42/unit ■■</span>
</div>
<div class="list-item">
<span class="supplier-name">Precision Parts Co.</span>
<span><span class="scrap-badge low-scrap">Scrap 2.1%</span> · Cost $56/unit</span>
</div>
<div class="list-item">
<span class="supplier-name">ValueSource India</span>
<span><span class="scrap-badge" style="background:#fce4e4;">Scrap 7.3%</span> · Cost $39/unit ■</span>
</div>
<div class="list-item">
<span class="supplier-name"><i class="fas fa-check-circle" style="color:#0e6245;"></i> Stellar
Motors</span>
<span><span class="scrap-badge low-scrap">Scrap 1.8%</span> · Cost $52/unit</span>
</div>
</div>
<div style="margin-top: 18px; background: #eaf3ff; border-radius: 16px; padding: 12px 14px;">
<span style="font-weight:600;"><i class="fas fa-clipboard-list"></i> Insight:</span> Some suppliers provide
lower-cost materials but create more scrap. Best: Axon (low scrap + competitive cost).
</div>
</div>
</div>
<!-- QUALITY IMPROVEMENT ROADMAP & TARGETS -->
<div style="margin-top: 20px;">
<div class="chart-card">
<div class="card-header">
<h3><i class="fas fa-road"></i> Quality Improvement Roadmap & Targets</h3>
<span class="target-tag" style="background:#e1f5fe; color:#0369a1;">Six Sigma 2026</span>
</div>
<div class="roadmap-grid">
<div>
<ul class="target-list">
<li><i class="fas fa-chart-line"></i> Reduce Defect Rate <span class="progress-badge">Target <2%< /span>
</li>
<li><i class="fas fa-stopwatch"></i> Reduce Downtime <span class="progress-badge">-18% Q2</span></li>
<li><i class="fas fa-handshake"></i> Improve Supplier Quality <span class="progress-badge">Audit
Q2</span></li>
<li><i class="fas fa-chalkboard-user"></i> Train Workers & Supervisors <span class="progress-badge">80%
certified</span></li>
<li><i class="fas fa-ban"></i> Use only suppliers with low scrap rate <span class="progress-badge">
<3% scrap</span>
</li>
</ul>
</div>
<div class="six-sigma-box">
<h4><i class="fas fa-chart-pie"></i> Six Sigma progress</h4>
<div style="font-size: 32px; font-weight: 600; margin: 10px 0;">3.8 σ</div>
<div style="display:flex; gap: 8px; margin-bottom: 16px;">
<span style="background:#2e5a9c; padding:2px 10px; border-radius: 20px;">DPMO 9,200</span>
<span style="background:#2e5a9c; padding:2px 10px; border-radius: 20px;">Goal 3.4 DPMO</span>
</div>
<p style="opacity:0.9; font-size:14px;">Focus on reducing defects, lowering downtime, improving supplier
quality, and training. Achieving targets will reach Six Sigma.</p>
<p style="margin-top:12px;"><i class="fas fa-check-circle"></i> Roadmap: Q2 audits · Q3 training · Q4
certification</p>
</div>
</div>
<!-- additional target reminder -->
<div style="display: flex; gap: 12px; margin-top: 24px; flex-wrap: wrap;">
<div style="background: #f6f9fc; border-radius: 30px; padding: 8px 18px;"><i class="fas fa-circle-check"
```

```
style="color:#1e7e34;"></i> Defect Rate: 3.2% → target 1.8%</div>
<div style="background: #f6f9fc; border-radius: 30px; padding: 8px 18px;"><i class="fas fa-circle-check"
style="color:#1e7e34;"></i> Downtime reduction: -12% achieved</div>
<div style="background: #f6f9fc; border-radius: 30px; padding: 8px 18px;"><i class="fas fa-circle-check"
style="color:#1e7e34;"></i> Supplier scrap threshold 3%</div>
</div>
</div>
</div>
<!-- footnote with same mention as screenshot -->
<div class="watermark">
<i class="far fa-clipboard"></i> SwiftMotors Quality Review • Cognos Analytics • April 2026 • Data refreshed
04/10/2026
</div>
</div>
<script>
(function () {
// ---------- PLANT OUTPUT CHART (bar) ----------
const ctxPlant = document.getElementById('plantOutputChart').getContext('2d');
new Chart(ctxPlant, {
type: 'bar',
data: {
labels: ['Plant A (Pune)', 'Plant B (Gurugram)', 'Plant C (Chennai)', 'Plant D (Sanand)'],
datasets: [{
label: 'Units Produced (Apr 2026)',
data: [3120, 2780, 4280, 1760],
backgroundColor: ['#1f77b4', '#ff7f0e', '#2ca02c', '#9467bd'],
borderRadius: 8,
barPercentage: 0.65,
}]
},
options: {
responsive: true,
maintainAspectRatio: false,
plugins: {
legend: { display: false },
tooltip: { backgroundColor: '#0b1c3a' }
},
scales: {
y: { beginAtZero: true, grid: { color: '#e4e9f0' }, title: { display: true, text: 'Units' } },
x: { grid: { display: false } }
}
}
});
// ---------- DEFECT RATE by PLANT & SHIFT (grouped bar) ----------
const ctxDefect = document.getElementById('defectShiftChart').getContext('2d');
new Chart(ctxDefect, {
type: 'bar',
data: {
labels: ['Plant A', 'Plant B', 'Plant C', 'Plant D'],
datasets: [
{
label: 'Day Shift',
data: [2.3, 3.1, 1.9, 2.5],
backgroundColor: '#5470a6',
borderRadius: 6,
},
{
label: 'High/Night Shift',
data: [3.8, 4.5, 3.2, 4.0],
backgroundColor: '#d95f5f',
borderRadius: 6,
}
]
},
options: {
responsive: true,
maintainAspectRatio: false,
plugins: {
tooltip: { callbacks: { label: (ctx) => `${ctx.dataset.label}: ${ctx.raw}%` } },
legend: { position: 'top', labels: { usePointStyle: true, boxWidth: 10 } }
},
scales: {
y: {
beginAtZero: true,
max: 6,
grid: { color: '#e0e6ed' },
title: { display: true, text: 'Defect rate (%)' }
},
```

```
x: { grid: { display: false } }
}
}
});
// ---------- DEFECT PARETO / TOP DEFECT TYPES (bar chart to represent categories) ----------
const ctxPareto = document.getElementById('defectParetoChart').getContext('2d');
new Chart(ctxPareto, {
type: 'bar',
data: {
labels: ['Network issues', 'Poor maintenance', 'Assembly quality', 'Material variance', 'Calibration'],
datasets: [{
label: 'Defect occurrences (%)',
data: [31, 24, 20, 15, 10],
backgroundColor: '#3b7cbf',
borderRadius: 6,
}]
},
options: {
indexAxis: 'y', // horizontal bar to look like Pareto / root cause
responsive: true,
maintainAspectRatio: false,
plugins: {
legend: { display: false },
tooltip: { callbacks: { label: (ctx) => `${ctx.raw}% of defects` } }
},
scales: {
x: { max: 40, grid: { color: '#dbe2ec' }, title: { display: true, text: 'Percentage (%)' } },
y: { grid: { display: false } }
}
}
});
// Additional small chart? we already have enough.
// also insert "Break Down % 4.34%" is already in KPI.
})();
</script>
<!-- small style tweaks to match cognos feeling -->
<style>
.kpi-card .kpi-value {
letter-spacing: -0.5px;
}
.chart-card {
transition: box-shadow 0.2s;
}
.chart-card:hover {
box-shadow: 0 12px 26px rgba(0, 20, 50, 0.06);
}
.fa-regular,
.fas,
.far {
pointer-events: none;
}
.list-item .scrap-badge {
font-weight: 500;
}
.six-sigma-box {
box-shadow: inset 0 0 0 1px #2a4870;
}
i {
color: #2c577c;
}
.badge-target,
.target-tag {
font-weight: 600;
}
</style>
</body>
</html>
```

■ 5. Results and Findings

5.1 Key Operational Metrics

4.34%	187 hrs	27%	5.4 / 10
Overall Defect Rate	**Downtime Hours**	**Re-admission / Rework**	**Avg Satisfaction**
Target < 2%	All Assembly Lines	Rework Required Rate	Quality Compliance Score

5.2 Critical Findings

• Defect Rate of 4.34% significantly exceeds the Six Sigma target of < 2%, with Ahmedabad plant showing 9.4% — the highest among all plants.

• Night shift consistently shows the highest defect rate across all plants. Combined with Afternoon shift, they account for 70% of all defects.

• Supervisor Prakash S has the highest defect rate at 6.13% — requiring targeted quality training intervention.

• Assembly LINE-7 (5.4%) and LINE-8 (5.0%) show highest per-line defect rates — both requiring process audits.

• GlassMart and SteelCo suppliers contribute the most scrap units (68 and 62 respectively) — contract review recommended.

• Negative correlation confirmed between material cost and defect rate — higher quality materials reduce defects.

5.3 Plant Performance Summary

Metric	Pune	Chennai	Ahmedabad	Overall
Best performing	✓	★ Leader		Chennai
Defect Rate	5.2%	3.4%	9.4%	4.34%
Units Produced	1,250	5,030	5,300	11,580
Defective Units	25	195	1,000	1,220
Status	**HIGH**	**MEDIUM**	**CRITICAL**	**HIGH**

■ 6. Conclusions and Recommendations

6.1 Conclusions

This project successfully demonstrated the power of IBM Cognos Analytics in transforming raw manufacturing data into actionable business intelligence. Through the implementation of 6 analytical deliverables, a comprehensive view of SwiftMotors' production quality was achieved.

The analysis confirms that SwiftMotors faces three interconnected challenges: high defect rates particularly in the Ahmedabad plant and Night shift, uneven supplier quality contributing to scrap costs, and specific assembly lines and supervisors consistently associated with higher defects. These findings provide a clear roadmap for achieving Six Sigma quality standards.

6.2 Recommendations

#	Action	Current State	Target
1	Reduce Defect Rate	4.34% overall / 9.4% Ahmedabad	< 2% (Six Sigma)
2	Night Shift Quality Control	4.64% defect rate	< 2% across all shifts
3	Supervisor Training	Prakash S: 6.13% defect rate	< 2% per supervisor
4	Assembly Line Audit	LINE-7: 5.4% & LINE-8: 5.0%	< 2% per line
5	Supplier Contract Review	GlassMart: 68 scrap units	< 20 scrap units
6	Reduce Downtime	187 hrs total downtime	< 100 hrs / quarter

6.3 Future Work

- Implement real-time dashboard monitoring with live production data feeds from assembly lines
- Expand dataset to include all 500+ assembly units and complete 2024 production year data
- Add predictive analytics for defect risk scoring using machine learning models
- Integrate shift scheduling data for automated supervisor workload optimization
- Build automated alerts when defect rate exceeds 2% threshold on any assembly line

■ 7. Appendix

A. Project Timeline

Phase	Activity	Date
Phase 1	KT Sessions — IBM Cognos Analytics training	Completed March 2026
Phase 2 Start	Project phase begins — Case Study 6 assigned	27 March 2026
Certification	IBM CEP Cognos Analytics certification	6 April 2026
Project Deadline	Case study submission	8 April 2026
Internship End	All deliverables submitted	10 April 2026

B. Tools and Technologies

Platform	IBM Cognos Analytics (ap2.ca.analytics.ibm.com)
Dataset Format	Microsoft Excel (.xlsx) — CS06_Automobile_SwiftMotors.xlsx
Report Module	IBM Cognos Report Studio — Q1, Q2, Q3
Exploration Module	IBM Cognos Exploration — Q4
Dashboard Module	IBM Cognos Dashboard — Q5
Story Module	IBM Cognos Story Making — Q6
Web Tech	HTML5, CSS3, Chart.js 4.4.0, IBM Plex Sans Font
Deployment	GitHub Pages — viacommand.github.io
Export Format	PDF for board presentation, HTML for web

C. Dataset Statistics

Total Records	200 vehicle production records
Total Columns	20 data fields (including calculated)
Plants	Pune, Chennai, Ahmedabad
Assembly Lines	LINE-1 through LINE-8 (8 lines)
Supervisors	Anand P, Prakash S, Rajesh K, Suresh M, Vijay R
Raw Material Suppliers	AutoParts India, ElecComp, GlassMart, PlastiTech, SteelCo
Vehicle Types	Sedan, SUV, EV (Electric Vehicle)
Defect Types	Assembly Gap, Electrical Fault, Engine Noise, Paint Defect, Trim Issue, Weld Failure
Date Range	January 2024 — December 2024
Defect Rate Range	0% to 9.4% across plants

End of Report — SwiftMotors Vehicle Manufacturing & Quality Analytics

IBM NM Internship Phase 2 | Submitted April 2026 | K Sam Giftson | Confidential

Final Submission (Repository + Deployed Link)

The final submission of the IBM Cognos Analytics Case Study Project includes both GitHub repositories and the live deployed website link for public access and evaluation.

1. Main Project Repository

This repository contains the HTML files and deployed web version of the project.

🔗 **Repository Link:**
https://github.com/viacommand/IBM-COGNOS-ANALYSTIC-PROJECT

🔗 **Live Deployed Link:**
https://viacommand.github.io/IBM-COGNOS-ANALYSTIC-PROJECT/

Repository Contents:
• index.html – Main landing page of the project
• Q5.HTML.html – Dashboard case study page
• Q6.HTML.html – Story Making case study page

2. Case Study Repository

This repository contains the complete IBM internship case study submission files and reports.

🔗 **Repository Link:**
https://github.com/viacommand/nm-ibm-intership-fe-case-study-6

Repository Contents:
• MY CASE 1.pdf – Production Quality Report
• MY CASE 2.pdf – Downtime Analysis Report
• MY CASE 3.pdf – Defect Type Distribution Report
• MY CASE 4.pdf – Exploration Analysis Report
• MY CASE 5.pdf – Manufacturing Operations Dashboard
• MY CASE 6.pdf – Cognos Story Making Report
• IBM_Cognos_Analytics_15_CaseStudy_Scenarios.pdf – Original case study document
• README.md – Project overview and documentation

Both repositories together provide the full source code, generated reports, documentation, and the deployed version of the IBM Cognos Analytics Case Study Project.

Thank you for giving me this wonderful opportunity to learn and showcase my skills through this project.